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CS2604.

Q1: Let  $V$  be a vector space and let  $v_1, v_2, v_3$  be linearly independent.

For  $a \in \mathbb{F}$ ,

$$\text{define } w_1 = v_1 + v_2 \quad w_2 = v_2 + v_3 \quad w_3 = v_3 + av_1.$$

Determine all values of  $a$  for which  $w_1, w_2, w_3$  form a basis of  $\text{span}(v_1, v_2, v_3)$ .

For  $w_1, w_2, w_3$  to be span of  $v_1, v_2, v_3$ , they have to be linearly independent.

So, for constant  $c_1, c_2, c_3 \in \mathbb{F}$ .

$$\Rightarrow c_1 w_1 + c_2 w_2 + c_3 w_3 = 0$$

$$\Rightarrow c_1(v_1 + v_2) + c_2(v_2 + v_3) + c_3(v_3 + av_1) = 0$$

$$\Rightarrow v_1(c_1 + ac_3) + v_2(c_1 + c_2) + v_3(c_2 + c_3) = 0$$

$$\left. \begin{array}{l} \Rightarrow c_1 + ac_3 = 0 \\ \Rightarrow c_1 + c_2 = 0 \\ \Rightarrow c_2 + c_3 = 0 \end{array} \right\} \begin{array}{l} \Rightarrow c_3(1+a) = 0 \quad \text{--- (i)} \\ \Rightarrow c_1 = -c_2 \\ \Rightarrow c_2 = -c_3 \end{array} \quad \left. \begin{array}{l} \vdots v_1, v_2, v_3 \neq 0 \\ \text{and} \end{array} \right\}$$

$$\text{for (eq. i)} \quad c_3(1+a) = 0$$

we have two case i)  $c_3 = 0$  OR ii)  $(1+a) = 0$

for eq. i to be 0. But  $c_3 = 0$  is ~~not~~ always true as  $v_1, v_2, v_3$  are LI

But if  $(1+a) = 0$ , then for non zero  $c_3$ ,  $c_3(1+a) = 0$  is the vector  $v_1, v_2, v_3$  are not LI.

So for them to be a basis, we must ensure,  $a+1 \neq 0 \Rightarrow \boxed{a \neq -1}$ .

So for  $v_1, v_2, v_3$  be a basis we must ensure  $a \neq -1$ .

b. For exceptional values of  $a$ , determine  $\dim(\text{span}(w_1, w_2, w_3))$ .

For exceptional values of  $a$ , ie  $a = -1$

$$\text{we have, } w_1 = v_1 + v_2$$

$$w_2 = v_2 + v_3$$

$$w_3 = v_3 + (-1)v_1.$$

If we do,  $w_1 - w_2 + w_3$  is

$$\Rightarrow v_1 + v_2 - v_2 - v_3 + v_3 - v_1 = 0.$$

ie  $w_1, w_2, w_3$  are linear dependent, so  $\dim(\text{span}(w_1, w_2, w_3)) \neq 3$ .

lets check for  $w_1, w_2$ .

$$c_1 w_1 + c_2 w_2 = 0$$

$$\Rightarrow c_1(v_1 + v_2) + c_2(v_3 + v_2) = 0$$

$$\Rightarrow v_1(c_1) + v_2(c_1 + c_2) + v_3(c_2) = 0$$

$\Rightarrow c_1 = 0, c_2 = 0$ . So  $w_1$  and  $w_2$  are linearly independent.

$$\text{So } \dim(\text{span}(w_1, w_2)) = \dim(\text{span}(w_1, w_2, w_3)) = 2.$$

2. Let  $V = \mathcal{P}_4(\mathbb{R})$ . and consider the following:-

$$p_1(x) = 1+x, \quad p_2(x) = x+x^2, \quad p_3(x) = x^2+x^3,$$

$$p_4(x) = x^3+x^4, \quad p_5(x) = 1+x^4.$$

a) Determine whether the five polynomials form a basis of  $\mathcal{P}_4(\mathbb{R})$ .

We know, for basis, we need linearly independent set of vectors.

considering  $c_1, c_2, c_3, c_4, c_5 \in \mathbb{R}$ .

$$\Rightarrow c_1 p_1(x) + c_2 p_2(x) + c_3 p_3(x) + c_4 p_4(x) + c_5 p_5(x) = 0.$$

$$\Rightarrow c_1(1+x) + c_2(x+x^2) + c_3(x^2+x^3) + c_4(x^3+x^4) + c_5(1+x^4) = 0$$

$$\Rightarrow (c_1+c_5) + x(c_1+c_2) + x^2(c_2+c_3) + x^3(c_3+c_4) + x^4(c_4+c_5) = 0.$$

$$\Rightarrow \left. \begin{array}{l} c_1 + c_5 = 0 \Rightarrow c_1 = -c_5 \\ c_1 + c_2 = 0 \Rightarrow c_1 = -c_2 \end{array} \right\} c_5 = c_2 = c_1$$

$$\left. \begin{array}{l} c_3 + c_4 = 0 \Rightarrow c_3 = -c_4 \\ c_2 + c_3 = 0 \Rightarrow c_3 = -c_2 \end{array} \right\} c_2 = c_4 = c_5$$

$$c_4 + c_5 = 0 \Rightarrow 2c_4 = 0 \Rightarrow c_4 = 0 \text{ i.e. } \boxed{c_1 = c_2 = c_3 = c_4 = c_5 = 0}$$

So it's a trivial case, i.e. the vectors form a basis for  $\mathcal{P}_4(\mathbb{R})$ .

b) If they do not, find a maximal independent subsets.

Yes, they are a basis.

c) Show that the vectors we discard can nevertheless be recovered from the vectors you are kept.

Are linearly independent, i.e. we can't remove anything without reducing the dimension.

d) Give a basis of  $P_4(\mathbb{R})$  consisting of linear combinations of  $p_1, \dots, p_5$  but not necessarily consisting of members of the original list.

Any scaling of our basis vectors remains linearly dependent.

$$p_1'(x) = 2 + 2x; \quad p_2'(x) = 2x + 2x^2; \quad p_3'(x) = 3x^2 + 3x^3$$

$$p_4'(x) = 2x^3 + 2x^4; \quad p_5'(x) = \frac{1}{2} + \frac{x^4}{2}.$$

So,  $\{p_1', \dots, p_5'\} = B'$  is our new Basis.

3. Let  $B = \{v_1, v_2, v_3, v_4\}$  be a basis of a vector space  $V$ .

with defined,  $u_1 = v_1 + v_2$ ,  $u_2 = v_2 + v_3$ ,  $u_3 = v_3 + v_4$ .

a) Prove that  $u_1, u_2, u_3$  are linearly independent.

Consider  $c_1, c_2, c_3 \in \mathbb{F}$ .

$$\text{we know, } c_1 u_1 + c_2 u_2 + c_3 u_3 = 0$$

$$\Rightarrow c_1 v_1 + c_2 (v_1 + v_2) + c_3 (v_2 + v_3) + c_4 (v_3) = 0.$$

$$\Rightarrow \boxed{c_1 = 0} \Rightarrow c_2 + c_3 = 0 \Rightarrow c_2 = -c_3.$$

$$\Rightarrow c_1 + c_2 = 0 \Rightarrow \boxed{c_3 = 0} \Rightarrow \boxed{c_2 = 0}$$

$$\Rightarrow c_1 = -c_2$$

So,  $u_1, u_2, u_3$  are linear independent.

b) Find all vectors  $u_4 \in V$  for which  $u_1, u_2, u_3, u_4$  is a basis of  $V$ .

Let our new vector  $u_4$  be  $a\bar{v}_1 + b\bar{v}_2 + c\bar{v}_3 + d\bar{v}_4$ .

considering  $c_1, c_2, c_3, c_4 \in \mathbb{F}$ .

$$\Rightarrow c_1 u_1 + c_2 u_2 + c_3 u_3 + c_4 u_4 = 0.$$

$$\Rightarrow c_1 + a c_4 = 0 \Rightarrow c_1 = -a c_4.$$

$$\Rightarrow c_1 + c_2 + b c_4 = 0 \Rightarrow c_2 = (a-b) c_4.$$

$$\Rightarrow c_2 + c_3 + c c_4 = 0 \Rightarrow c_3 = (-a+b-c) c_4.$$

$$\Rightarrow c_3 + d c_4 = 0 \Rightarrow c_4 \times (-a+b-c+d) = 0.$$

Same logic, we need  $(-a+b-c+d) \neq 0$  as if it equals to 0 for a non zero value of  $c_4$  we will get our solutions.

$$\text{So, } -a+b-c+d \neq 0$$

$$\Rightarrow \boxed{b+d \neq a+c} \text{ for all } a, b, c, d \text{ we will get } u_4 \text{ such that}$$

$u_1, u_2, u_3, u_4$  forms a basis.

c) Among ~~four~~ possible choices, determine which  $u_4$  lie in  $\text{span}(v_1, v_2, v_3)$

For  $u_4$  to lie in  $\text{span}(v_1, v_2, v_3)$ , we can write

$$u_4 = a v_1 + b v_2 + c v_3.$$

Comparing the above eqn with part (b), we notice,

our  $d = 0$ , as we don't have any  $v_4$  part.

$$\Rightarrow -a+b-c+(0) \neq 0$$

$$\Rightarrow \boxed{b \neq a+c} \forall a, b, c \text{ we can guarantee } u_4 \in \text{span}(v_1, v_2, v_3).$$

d) Explain why part (b) is an useful application rather than mere computation.

In given problem, we have a vector space  $V$  where we atleast have 4 LI vectors  $(v_1, v_2, v_3, v_4)$  ie the  $\dim(V) = 4$ .

In part (a) we made a new set of LI vectors from L. Comb. of  $v_1, \dots, v_4$  as  $u_1, u_2, u_3$ , so  $\text{span}(u_1, u_2, u_3) = V'$  (let) where  $V' \subsetneq V$ .  
and  $\dim(V') = 3$ .

We found a new Linearly independent vector  $u_4$  from  $v_1, v_2, v_3, v_4$  and we add it in set  $u_1, u_2, u_3$  such that  $\text{span}(u_1, u_2, u_3, u_4) = V''$  (let)  
and  $\dim(V'') = 4$ .

So we ~~ex~~ 'EXTEND' our given BASIS' to a higher order/dim space.

We can conclude this for any given smaller set to form of a larger or full basis.

Q4. Suppose  $U, W$  are subspaces of a finite dim vector space  $V$  and  $\dim(U) = 7$ ,  $\dim(W) = 6$ ,  $\dim(V) = 10$ .

a) Smallest possible value of  $\dim(U+W)$ ?

When we add  $(U+W)$  our dim will be atleast the largest one out of  $U, W$ . ie 7.

Largest possible value of  $\dim(U+W)$ ?

As  $U$  and  $W$  are subspaces of  $V$  we can't get more than  $\dim(V)$  ie 10.

a) Smallest value of  $\dim(U \cap W)$ .

we know,

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

$$\text{So, } \dim(U \cap W) = \dim(U) + \dim(W) - \dim(U+W).$$

$$= 7+6 - (10) = \boxed{3}$$

largest value of  $\dim(U \cap W)$ .

$$\text{again, } \dim(U \cap W) = \dim(U) + \dim(W) - \dim(U+W)$$

$$= 7+6 - 7 = \boxed{6}$$

b) For extreme values, can these  $U \cap W$  exist?

Yes, for max intersection i.e 6, we can imagine as  $W \subset U$  or  $U \subset W$ . Then  $W \cap U = W$ .

for min intersection, i.e 3, we can imagine as for  $V$  we have 10 LI vectors, build  $U$  with first 7,  $W$  with remaining 3 and then (3 common) from  $U$ .

d) If we define  $U+W = V$ , find  $\dim(U \cap W)$ ?

CASE 1.

If we give full <sup>unique</sup> rank to vectors to  $U$  from  $V$  i.e  $U$  has 7 unique vectors. To generate/cover the whole space  $V$  we need to have the remaining 3 unique vectors from  $V$  to  $W$  and as  $\dim(W)=6$ , we need to have common 3 vectors from  $U$ .

$$\text{i.e } U \cap W = 3.$$

### CASE II

Similarly, if we give full unique set to  $W$  is 6,  $U$  must have 1 remaining vectors and as  $\dim(U) = 7$  we again need to have 3 common vectors, i.e.  $U \cap W = 3$ .

So in both of the cases  $\boxed{U \cap W = 3}$ .

5) Let  $U, W, Z$  be subspaces of  $V$ , and suppose  $V = U \oplus W$  and  $W \subseteq Z \subseteq U$ . Again  $U \cap Z = \{0\}$ .

a) Proof or disprove :  $V = U \oplus Z$ .

We need to show:  $\Rightarrow V = U + Z$      $\cdot \Rightarrow U \cap Z = \{0\}$ .

The second condition is already given,  $U \cap Z = \{0\}$ .

We have,  $V = U \oplus W$  i.e.  $V = U + W$

Let's say  $\bar{v} \in V$ ,  $\bar{u} \in U$ ,  $\bar{w} \in W$ .

$$\text{So, } \bar{v} = \bar{u} + \bar{w}.$$

Also,  $W \subseteq Z$ , so  $\bar{w} \in Z$ .

$$\Rightarrow \bar{v} = \bar{u} + \bar{w} \subseteq U + Z.$$

$\Rightarrow V \subseteq U + Z$ , But as  $U$  and  $Z$  are subspaces of  $V$  itself.

We can't get more than  $V$ .

$$\Rightarrow \boxed{V = U + Z} \text{ hence, } V = U \oplus Z. \checkmark$$

b) Additional conditions for  $Z$ ?

We saw above that  $U \cap Z = \{0\}$  mathematically forces the condition of  $Z = W$ . [ $\because$   $Z$  can't be diff bigger / contain unique vector other than  $v$ ].

So,  $Z = W \Rightarrow V = U \oplus Z$  to be true as  $V = U \oplus W$  is given.

6. Let  $T: V \rightarrow V$  be linear, and let  $U \subseteq V$  be a subspace satisfying  $T(\bar{u}) = \bar{u} \quad \forall \bar{u} \in U$ . Also  $T(\bar{v}) = 0 \quad \forall \bar{v} \in W$ . and  $V = U \oplus W$ .

a) Determine  $T(u+w)$  for arbitrary  $\bar{u} \in U, \bar{w} \in W$ .

We know, from the properties of linear transformation,

$$T(c\bar{a} + \bar{b}) = cT(\bar{a}) + T(\bar{b}).$$

$$\Rightarrow T(\bar{u} + \bar{w}) = T(\bar{u}) + T(\bar{w}) = \bar{u} + 0 = \bar{u} \in U.$$

b) Determine null  $T$ .

null of a linear transformation means set of all vector from input space (here  $V$ ) which get maps to  $\bar{0}$ .

Here as  $T(\bar{v}) = 0 \quad \forall \bar{v} \in W$ , null  $T = W$ .

c) Determine Range  $T$ .

Range ( $T$ ) means all vector set which retain its dimensions.

Here  $T(\bar{u}) = \bar{u}, \bar{u} \in U$ , Range ( $T$ ) =  $U$ .

d) Determine whether  $T$  is invertible.

We know, every linear map is a function, and a function is invertible if it is bijective, i.e. both one-one and onto.

But for it to be one-one we must show null  $T = \{0\}$ .

But in our case null  $T = W$ .

So until  $W = \{0\}$ ,  $T$  is not invertible.

e. Let Transformation  $T$  be,

$$T(\bar{u}) = 2\bar{u} \quad \bar{u} \in U \quad \text{and} \quad T(\bar{w}) = -\bar{w} \quad \bar{w} \in W$$

Determine the null space and range.

Null space: Let  $v = u + w$ .

$$T(v) = T(u + w) = T(u) + T(w)$$

$$\Rightarrow 2u - w \quad \text{let this } T(v) = 0$$

$$\text{ie } 2u - w = 0$$

$$\Rightarrow 2u = w \quad [\because \bar{u}, \bar{w} \text{ are vectors, a twice of a vector equals another vector means } \bar{0}].$$

$$\text{So, } u = w = 0 \quad \text{so } v = 0 \quad \text{so } \text{Null } T = \{0\}.$$

Range space: So  $T(u/2) = u$  and  $T(-w) = w$ .

we find any arbitrary vector  $v = u + w$  by  $T(u/2 - w) = u + w$ .

So  $\text{Range } T = \text{whole space ie } V$ .

f. Prove  $T^{-1}$  is feasible OR  $T$  is invertible.

M1 Given,  $T(\bar{u}) = 2\bar{u}$  and  $T(\bar{w}) = -\bar{w}$

$$\Rightarrow T^{-1}T(\bar{u}) = T^{-1}(2\bar{u}) \quad \Rightarrow T^{-1}T(\bar{w}) = T^{-1}(-\bar{w})$$

$$\Rightarrow \bar{u} = 2T^{-1}(\bar{u}) \quad \Rightarrow \bar{w} = -T^{-1}(\bar{w})$$

$$\Rightarrow T^{-1}(\bar{u}) = \bar{u}/2 \quad \Rightarrow T^{-1}(\bar{w}) = -\bar{w}$$

As,  $T^{-1}$  is existing, it is possible.

M2  $T$  is invertible if it is a bijection, i.e.

• into : Yes as  $\text{null } T = \{0\}$  (shown in (e)).

• onto : Yes as  $\text{Range } T = V$  (shown in (e))

hence,  $T^{-1}$  is bijection, i.e. invertible.

7. Let  $T: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$  defined by,

$$T(p) = p + p' \text{ where } p \in \mathcal{P}_3(\mathbb{R}) \text{ and } p' \text{ is first derivative of } p.$$

a) Prove that  $T$  is linear.

For linear, we need to show,

$$T(cp_1 + p_2) = cT(p_1) + T(p_2), \text{ where } p_1, p_2 \in \mathcal{P}_3(\mathbb{R}).$$

Let our Basis for  $\mathcal{P}_3(\mathbb{R}) = \{1, x, x^2, x^3\}$  is standard basis.

$$\text{So, let } p_1 = a_1 + b_1x + c_1x^2 + d_1x^3.$$

$$p_2 = a_2 + b_2x + c_2x^2 + d_2x^3.$$

$$\text{LHS, } T(cp_1 + p_2) = T(ca_1 + cb_1x + cc_1x^2 + cd_1x^3 + a_2 + b_2x + c_2x^2 + d_2x^3).$$

$$= T(ca_1 + a_2 + (cb_1 + b_2)x + (cc_1 + c_2)x^2 + (cd_1 + d_2)x^3)$$

$$= ca_1 + a_2 + (cb_1 + b_2)x + (cc_1 + c_2)x^2 + (cd_1 + d_2)x^3 + (cb_1 + b_2)x + 2(cc_1 + c_2)x + 3(cd_1 + d_2)x^2.$$

$$= (ca_1 + cb_1 + a_2 + b_2) + (cb_1 + 2cc_1 + b_2 + 2c_2)x + (cc_1 + 3cd_1 + c_2 + 3d_2)x^2 + (cd_1 + d_2)x^3. \quad \text{--- (i).}$$

$$\text{RHS, } cT(p_1) + T(p_2).$$

$$= cT(a_1 + b_1x + c_1x^2 + d_1x^3) + T(a_2 + b_2x + c_2x^2 + d_2x^3).$$

$$= c\{a_1 + b_1x + c_1x^2 + d_1x^3 + b_1 + 2c_1x + 3d_1x^2\} + (a_2 + b_2x + c_2x^2 + d_2x^3 + b_2 + 2c_2x + 3d_2x^2)$$

$$= ca_1 + cb_1 + a_2 + b_2 + (cb_1 + 2cc_1 + b_2 + 2c_2)x + (cc_1 + 3cd_1 + c_2 + 3d_2)x^2 + (cd_1 + d_2)x^3 \quad \text{--- (ii)}$$

As we have  $\text{LHS} = \text{RHS}$ , we conclude it as linear.

b) Determine whether the subspace  $U = P_1(\mathbb{R})$  is mapped into itself by  $T$ .

We know, into condition is a transformation  $T: V_1 \rightarrow V_2$  when a vector  $\vec{v} \in V_1$  and  $T(\vec{v}) \in V_2$ .

Checking  $T: U \rightarrow U$ .

considering a basis  $B = \{1, x\}$  for  $P_1(\mathbb{R})$

a vector  $\vec{u} = a + bx \in U$ .

$$T(\vec{u}) = a + bx + b = a + b + bx \Rightarrow c + bx \text{ if } c = a + b$$

is in the form of  $B$ . So  $T(\vec{u}) \in U$  - ie into.

c) Determine whether  $U$  is mapped onto itself.

Onto means any  $\vec{v}$  vector has a representation let  $\vec{u} \mid T(\vec{u}) = \vec{v}$   
 $\vec{v}, \vec{u} \in U$ .

let our  $\vec{v} = c + dx$ . and let our initial vector  $= \vec{u} = a + bx$ .

$$\text{So, } T(\vec{u}) = \vec{v}$$

$$\Rightarrow T(a + bx) = a + bx + b = c + dx$$

$$= a + b + bx = c + dx.$$

$$\Rightarrow a + b = c \Rightarrow a = c - b = c - d.$$

$$b = d$$

$$\therefore \vec{u} = (c - d) + dx \xrightarrow{T} c + dx.$$

So  $T$  is onto also.

d) Determine the null  $T$ .

we know  $T: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$

Basis for  $\mathcal{P}_3(\mathbb{R})$  is  $\{1, x, x^2, x^3\}$ .

For null space we know,  $T(\bar{x}) = 0 \quad \forall x \in V$ .

Let  $\bar{x} = a + bx + cx^2 + dx^3$ .

$$T(\bar{x}) = a + bx + cx^2 + dx^3 + b + 2cx + 3dx^2 = 0$$

$$\Rightarrow a + b + x(b + 2c) + x^2(c + 3d) + x^3(d) = 0$$

$$\Rightarrow a + b = 0 \quad \Rightarrow a = 0$$

$$\Rightarrow b + 2c = 0 \quad \Rightarrow b = 0$$

$$\Rightarrow c + 3d = 0 \quad \Rightarrow c = 0 \quad \Rightarrow \bar{x} = \bar{0}$$

$$\Rightarrow d = 0$$

So  $\exists$  only vector  $\bar{0}$  which get reduce to  $\bar{0}$

So  $\text{Null } T = \{\bar{0}\}$ .

e) Determine whether  $T$  is invertible or not.

As we proved  $\text{Null } T = \{0\} \Rightarrow \dim(\text{Null } T) = 0$ .

we know,  $\dim(V) = \dim(\text{Null } T) + \dim(\text{Range } T)$ .

$$\Rightarrow 4 = 0 + \dim(\text{Range } T) = 4.$$

So,  $\dim(\text{Null } T) = 0 \Rightarrow$  into  $\left. \begin{array}{l} \dim(\text{Range } T) = 4 \text{ (full)} \Rightarrow \text{onto.} \end{array} \right\} \rightarrow \text{bijection} \rightarrow T \text{ is invertible.}$

f) If  $q \in \mathcal{P}_3(\mathbb{R})$ , determine whether there exist a unique  $p \in \mathcal{P}_3(\mathbb{R})$  satisfying  $p + p' = q$ .

Let there be two polynomials  $p_1, p_2$  where.

$$p_1 + p_1' = q \quad \text{and}$$

$$p_2 + p_2' = q.$$

$$\Rightarrow p_1 + p_1' = p_2 + p_2'$$

$$\Rightarrow (p_1 - p_2) + (p_1' - p_2') = 0$$

$$\text{Let } p_1 - p_2 = h \Rightarrow h' = p_1' - p_2'$$

$$\Rightarrow h + h' = 0$$

from part e we show, if  $T(\bar{x}) = 0 \Rightarrow \bar{x} = 0$  as  $\text{null } T = \{0\}$ .

$$\therefore h = 0 \Rightarrow p_1 = p_2$$

So  $\exists$  only a unique  $p$  for which  $T(p) = q$ .

8. Let  $T: V \rightarrow V$  be a linear operator on an  $n$ -dim vector space and  $\dim(\text{null } T) = k$ .

a) what is the largest possible <sup>dim</sup> sub-space  $U \subseteq V$  on ~~for~~ which  $T$  is one to one?

$T: U \rightarrow U$ , for  $T$  to be one to one, we know  $\text{null } T = \{0\}$

So  $U$  can't include any dimension from  $\text{null}(T)$ .

$$\text{So } \dim(U) = n - k \text{ (max).}$$

b. Show that such a sub-space always exists.

c. Can  $U$  be chosen such that,  $V = \text{null } T \oplus U$ .

If we proved  $U \oplus \text{null } T = V$  where always exists we automatically show  $\exists U$  where  $\dim(U) = n - k$ .

Yes,  $U$  can always be chosen such that  $V = \text{null } T \oplus U$ .

• Let  $\dim(\text{null } T) = k \rightarrow$  Basis of the null space

$$B_1 = \{v_1, \dots, v_k\}.$$

• By basis extension theorem, any finite linearly independent list can be extended to a basis of a finite dimensional vector space, by adding more linearly independent vectors.

Let Basis  $B_2 = \{v_1, \dots, v_k, u_1, \dots, u_{n-k}\}$  for  $V$ .

• Let us define  $U = \text{span}(u_1, \dots, u_{n-k})$ .

• As the set  $B_2$  is basis for  $V$ , the combination of two span generates  $V$  as  $V = \text{span}(v_1, \dots, v_k) + \text{span}(u_1, \dots, u_{n-k})$ .

$$\Rightarrow V = \text{null } T + U. \quad - (i)$$

• And because it is a basis, the vectors are strictly linearly independent in  $\text{null } T \cap U = \{0\}$  - (ii)

from eq. i, ii we conclude  $V = \text{null } T \oplus U = V$ .

d) If  $k > 0$ , can  $T: U \rightarrow V$  ever be onto?

Given,  $k > 0$ . we know  $\max \dim(U) = n - k$ .

We know  $T$  is a function from  $U$  to  $V$ .

$$U \xrightarrow{T} V$$

where  $\dim(U) = n - k$  and  $\dim(V) = n$ .

As  $k > 0 \Rightarrow n - k < n$ .

But for onto, from  $A \rightarrow B$  we need  $|A| \geq |B|$ .

Here we don't have  $\dim(U) \geq \dim(V)$ .

So  $T: U \rightarrow V$  is never onto if  $k > 0$ .

e) If  $k = 0$ , what structural conclusion can you draw about  $T$ .

If  $k = 0 \Rightarrow \dim(\text{null } T) = 0$  i.e.  $\text{null } T = \{0\}$ .

We know, if a linear map  $T$  has  $\text{null } T = \{0\}$ .

it means,  $T$  is into. (one to one).

And if a linear map  $T$  maps from finite dim to finite dim vector space and  $T$  is into it means  $T$  is onto.

And if  $T$  is into and onto  $\rightarrow$  bijection or invertible.

$\therefore T: U \rightarrow V$  is invertible  $\Rightarrow$

$$\dim(U) = \dim(V)$$

$$n - k = n \Rightarrow \boxed{k = 0} \text{ Given condition.}$$

9. Let  $U, W$  be a finite-dimensional sub-spaces of  $V$ .  
 Suppose  $\dim(U) = r$ ,  $\dim(W) = s$ ,  $\dim(U \cap W) = k$ .

a) Prove that there exists a basis of  $V$  of the form.

$$u_1, \dots, u_k, u_{k+1}, \dots, u_r, w_{k+1}, \dots, w_s, z_1, \dots, z_m.$$

where

$u_1, \dots, u_k$  is a basis of  $U \cap W$ ,

$u_1, \dots, u_r$  is a basis of  $U$  and

$u_1, \dots, u_k, w_{k+1}, \dots, w_s$  is a basis of  $W$ .

Proof:-

• Let  $U \cap W$  be a subspace of  $V$ . Let  $\{u_1, \dots, u_k\}$  be a basis for  $U \cap W$ .

• Extending to  $U$ , because  $U \cap W \subseteq U$ , the set  $\{u_1, \dots, u_k\}$  is linearly independent. By the basis extension theorem, we can append  $(r-k)$  vectors to form basis of  $U$ :  $\{u_1, \dots, u_k, u_{k+1}, \dots, u_r\}$ .

• Extending to  $W$ , similarly  $U \cap W \subseteq W$ , we can extend our set  $\{u_1, \dots, u_k\}$  to form basis of  $W$  by,  $\{u_1, \dots, u_k, w_{k+1}, \dots, w_s\}$ .

• Is our new set, i.e. mixture of vectors from  $U, W$  linearly independent?

Assuming a linear combination over field  $F$ :

$$c_1 \bar{u}_1 + \dots + c_r \bar{u}_r + d_{k+1} \bar{w}_{k+1} + \dots + d_s \bar{w}_s = 0.$$

$$\Rightarrow c_1 \bar{u}_1 + \dots + c_r \bar{u}_r = - (d_{k+1} \bar{w}_{k+1} + \dots + d_s \bar{w}_s)$$

Here our LHS vector (let  $\bar{u}'$ ) is completely in  $U$  and RHS vector (let  $\bar{v}'$ ) completely in  $W$ .

But  $\bar{u}' = \bar{v}'$

So,  $\bar{u}' / \bar{v}' \in \mathcal{U} \cap \mathcal{W}$ .

Hence we can write  $\bar{u}' / \bar{v}'$  solely as a combination of Basis( $\mathcal{U} \cap \mathcal{W}$ )

$$\Rightarrow -(d_{k+1}\bar{w}_{k+1} + \dots + d_s\bar{w}_s) = a_1\bar{u}_1 + \dots + a_k\bar{u}_k.$$

$$\Rightarrow a_1\bar{u}_1 + \dots + a_k\bar{u}_k + d_{k+1}\bar{w}_{k+1} + \dots + d_s\bar{w}_s = 0.$$

So As,  $\bar{u}_1, \dots, \bar{u}_k$  is a basis for  $\mathcal{U} \cap \mathcal{W} \Rightarrow a_1 = \dots = a_k = 0$ .

This forces  $d_1 = \dots = d_s = 0$ .

$$\text{Similarly, } (c_1\bar{u}_1 + \dots + c_r\bar{u}_r) = a_1\bar{u}_1 + \dots + a_k\bar{u}_k$$

$$\Rightarrow a_1 = \dots = a_k = 0 \Rightarrow c_1\bar{u}_1 + \dots + c_r\bar{u}_r = 0$$

As,  $(u_1, \dots, u_r)$  is a basis for  $\mathcal{U}$ ;  $\Rightarrow c_1 = \dots = c_r = 0$

So, our set is linearly independent.

• Extending to  $\mathcal{V}$ , we made  $\mathcal{U} \cup \mathcal{W}$  now we should add the remaining vectors (let  $m$ ) to our set to completely our set of Basis for  $\mathcal{V}$ ,

$$\text{Basis}(\mathcal{V}) = \{u_1, \dots, u_k, u_{k+1}, \dots, u_r, w_{k+1}, \dots, w_s, z_1, \dots, z_m\}.$$

b) Using above proof to prove  $\therefore \dim(U+W) = r+s-k$ .

By definition, the subspace  $U+W$  is spanned by the union of any basis of  $U$  and any basis of  $W$ .

Using the bases defined in part (a), i.e.  $S = \{u_1, \dots, u_r, w_{k+1}, \dots, w_s\}$

we also showed  $S$  is linearly independent.

Because  $S$  spans  $U+W$  and is linear independent,  $S$  is a basis for  $U+W$ .

The dimension is simply the total no. of vectors in  $S$ .

$\rightarrow$  There are  $r$  vectors in the sequence  $u_1, \dots, u_r$ .

$\rightarrow$  There are  $(s-k)$  vectors in the sequence  $w_{k+1}, \dots, w_s$ .

$$\therefore \dim(U+W) = r + (s-k) = r+s-k.$$

c) Explain exactly where the basis-extension theorem is being used.

In part (a), we have used basis extension theorem thrice

- Extending the linear independent set  $\{u_1, \dots, u_k\}$  to form the basis for  $U$ .
- Extending that same set  $\{u_1, \dots, u_k\}$  to form the basis for  $U$ .
- Extending the combined linearly independent set  $\{u_1, \dots, u_r, u_{k+1}, \dots, w_s\}$  to form the final basis for  $V$ .

d. Deduce the criterion for  $U+W = U \oplus W$ .

By definition, the sum of two sub-space  $U+W$  is a direct sum is  $U \oplus W$  iff their intersection is trivial i.e.  $U \cap W = \{0\}$ .

If  $U \cap W = \{0\}$ , then  $\dim(U \cap W) = 0$ .

So,  $K=0$ .

Hence,  
~~So~~  $\dim(U+W) = r+s \Rightarrow U \oplus W = V$ .

10. Let  $V$  be a 6 dimensional vector space.

We want to construct a linear operator  $T: V \rightarrow V$

satisfying all of the following:

-  $\dim(\text{null } T) = 2$  -  $\dim(\text{Range } T) = 4$  and  $\dim(\text{null } T \cap \text{Range } T) = 1$ .

a) Prove that such operator can exist.

By the rank-Nullity Theorem, a linear operator  $T: V \rightarrow V$ .

on finite dim. space  $\dim(\text{null } T) + \dim(\text{Range } T) = \dim(V)$ .

Our given data,  $\dim(\text{null } T) + \dim(\text{Range } T) = \dim(V) \Rightarrow 2+4=6$ .

Also  $\dim(\text{null } T \cap \text{Range } T) \leq \min(\overset{\dim}{\text{null } T}, \dim \text{Range } T)$ .

and we have  $1 \leq \min(2, 4)$ .

So, as there are not structural contradictions, such an operator can exist.

b) Construct one explicitly by choosing a basis  $= \{v_1, v_2, v_3, v_4, v_5, v_6\}$ .

Given, Basis of  $T = \{v_1, v_2, v_3, v_4, v_5, v_6\}$ .

$\begin{aligned} & \cdot T(v_1) = 0 \\ & \cdot T(v_2) = 0 \end{aligned} \left. \vphantom{\begin{aligned} & \cdot T(v_1) = 0 \\ & \cdot T(v_2) = 0 \end{aligned}} \right\} \dim(\text{null}(T)) = 2.$

$\cdot T(v_3) = v_1 \left. \vphantom{\cdot T(v_3) = v_1} \right\} \dim(\text{null } T \cap \text{Range } T) = 1$

$\cdot T(v_4) = v_4$

$\cdot T(v_5) = v_5$

$\cdot T(v_6) = v_6$

$\left. \vphantom{\begin{aligned} & \cdot T(v_4) = v_4 \\ & \cdot T(v_5) = v_5 \\ & \cdot T(v_6) = v_6 \end{aligned}} \right\} \dim(\text{Range } T) = 4.$

c) verify all three properties required.

From our construction in part (b).

1.  $\dim(\text{Range } T) = 4.$

The range is the span of the outputs:  $\text{span}(0, 0, v_1, v_4, v_5, v_6).$   
is  $\text{span}(v_1, v_4, v_5, v_6)$ . As ~~the~~ these four vectors are part of our original basis, they are linearly independent.

2.  $\dim(\text{null } T) = 2.$

If we set  $T(c_1 v_1 + \dots + c_6 v_6) = 0$ , we get  $c_3 v_1 + c_4 v_4 + c_5 v_5 + c_6 v_6 = 0 \Rightarrow c_3 = c_4 = c_5 = c_6 = 0$  so  $c_1 v_1$  and  $c_2 v_2$  are left free to be anything. So the null space is exactly  $\text{span}(v_1, v_2)$  is  $\dim(\text{null } T) = 2.$

3.  $\dim(\text{null } T \cap \text{Range } T) = 1.$

The null space is generated by  $\{v_1, v_2\}$ . The range is generated by  $\{v_1, v_4, v_5, v_6\}$ . Only common vector is  $v_1$ .

So ~~dim~~  $\text{null } T \cap \text{Range } T = \text{span}\{v_1\}$  is dim of 1.

d) Construct a space with  $\text{null } T \cap \text{Range } T = \{0\}.$

• we have  $\dim(\text{null } T) = 2$ ,  $\dim(\text{Range } T) = 4.$

$$\begin{array}{l} \hookrightarrow T(v_1) = 0 \\ \hookrightarrow T(v_2) = 0 \end{array} \left. \begin{array}{l} \text{null } T \\ \text{span}(v_1, v_2) \end{array} \right\} \begin{array}{l} \hookrightarrow T(v_3) = v_3 \\ \hookrightarrow T(v_4) = v_4 \\ \hookrightarrow T(v_5) = v_5 \\ \hookrightarrow T(v_6) = v_6 \end{array} \left. \begin{array}{l} \text{Range } T \\ \text{span}(v_3, v_4, v_5, v_6) \end{array} \right\}$$

e) Explain why the data do not determine relative position?

The data given is,  $\dim(\text{null } T) = 2$  and  $\dim(\text{Range } T) = 4$ .

Only tells us about size of the sub-space.

As we shown in part (b) and (d) that any thing can be made  
is the intersection can be 1D/0D. is the intersection of  
the spaces are difficult to determine solely on the size of  
sub-spaces.

f) Determine whether either operator can be invertible.

A linear transformation on a finite-dim space is invertible  
iff it is a bijection (into + onto).

- For an map to be into (one-to-one) the null space =  $\{0\}$
- For a map to be onto (surjective) the  $\text{Range } T = \text{dim } V$ .

But  $\dim(\text{null } T) = 2$  is not 0 and

$\dim(\text{Range } T) = 4$  is not 6 so

$T$  will never be invertible.

4e) Suppose instead that  $U \cap W = \{0\}$  what can we conclude  
about whether  $U + W = V$ .

$$\text{If } U \cap W = \{0\} \Rightarrow \dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W) \\ \Rightarrow 10 \neq 6 + 7 + 0 = 13$$

So mathematically it is impossible to have  $U \cap W = \{0\}$ .